THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING

DEPARTMENT OF CIVIL ENGINEERING

CIVL1113: ENGINEERING MECHANICS AND MATERIALS

11 May 2015 TIME: 14:30 – 17:30 (3 hours)

Answer **5 (FIVE)** questions.
All questions carry equal marks.
Marks for sub-questions (if any) are given.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is the candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A

1. Beam *ABCDEF* (see Figure Q1) has a series of simple supports (pin and rollers) with hinges at points *C* and *D*. Vertical point loads, inclined point loads, uniformly distributed loads and triangularly distributed load are applied at various locations as shown in the same figure.
   (a) Determine the reactions (value and direction) at supports *A*, *B*, *E* and *F*. **(4 marks)**
   (b) Draw (on tension side) to scale the bending moment diagram (BMD) as well as draw to scale the shear force diagram (SFD). Mark on the BMD and SFD all the pertinent coordinates including the locations of zero moments. **(14 marks)**
   (c) Draw to scale the axial force diagram (AFD). Mark on the AFD all the pertinent values. **(2 marks)**

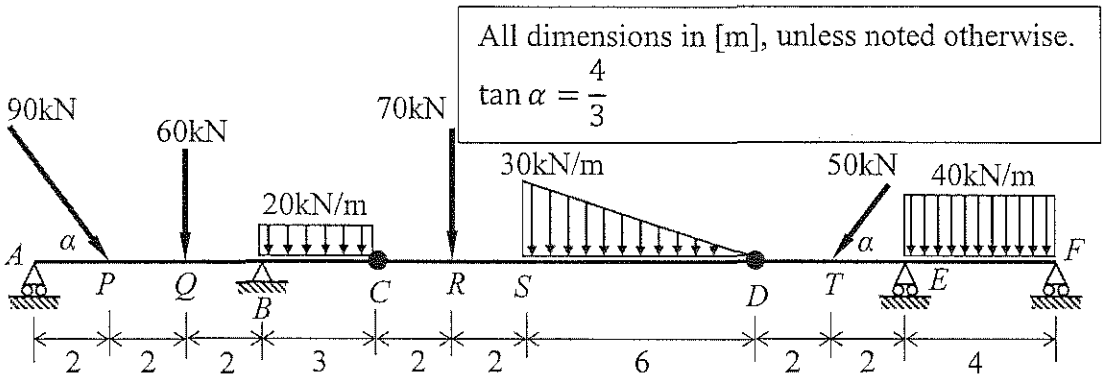


Figure Q1

2. (a) A simply supported composite beam *AB* with overhang is subjected to uniformly distributed load (UDL). The span length and the overhang are 7m and 3m, respectively. The composite beam consists of 2 different materials with identical cross-section dimensions, stacked as shown in Figure Q2a. Properties of the materials are as follows:
   Youngs modulus: Material 1: $E_1 = 450$GPa
   Material 2: $E_2 = 150$GPa
   Allowable bending stress in tension and compression: Material 1: $\sigma_1 = \pm 80$MPa
   Material 2: $\sigma_2 = \pm 40$MPa
   Calculate the maximum UDL w in kN/m that can be carried by the beam. **(14 marks)**

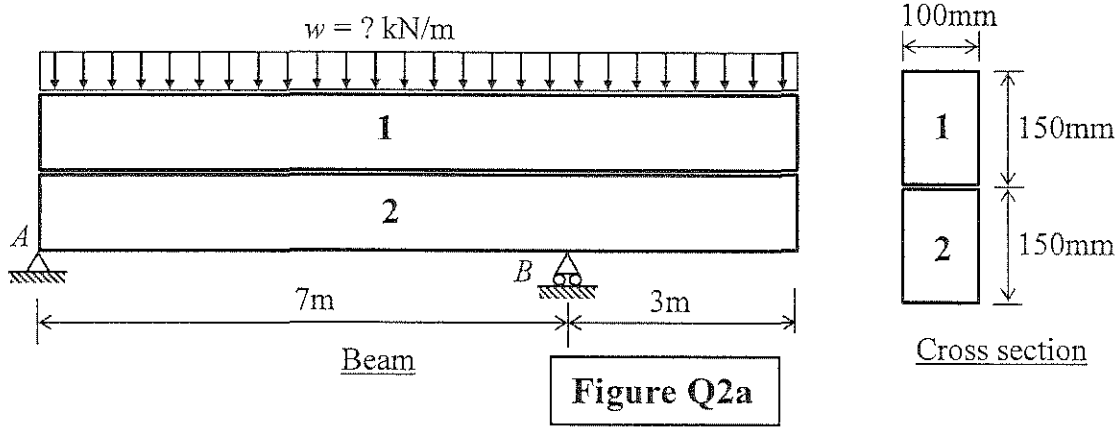


Figure Q2a

2. (b) Sketch the bending moment diagram (BMD) and shear force diagram (SFD) for the simply supported beam $ABCD$ shown in Figure Q2b, where $AB = BC = CD$ and $P_1 < P_2$. **(6 marks)**

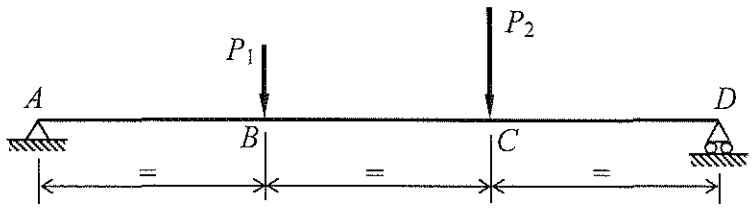
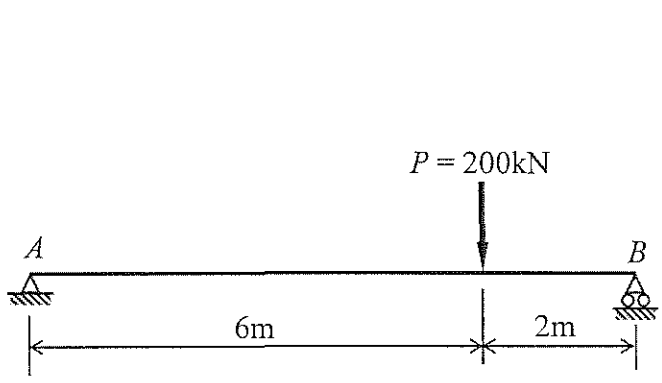


Figure Q2b

3. A simply supported beam AB with span of 8m is subjected to a point load P = 200kN at a distance 6m from A, as illustrated in Figure Q3. The beam has a made-up hollow rectangular cross section consisting of 4 pieces of timber plates connected by 8 rows of 12mm diameter bolts, as shown in the same figure. If the allowable average shear stress of each bolt is τ= 100MPa:
   - Suggest the longitudinal spacing of the bolts (must be in a round number multiple of 10mm).
   - If the first set of bolts must be installed within a distance of 50 to 100mm from the support, determine the total number of bolts needed in the beam.

   (20 marks)



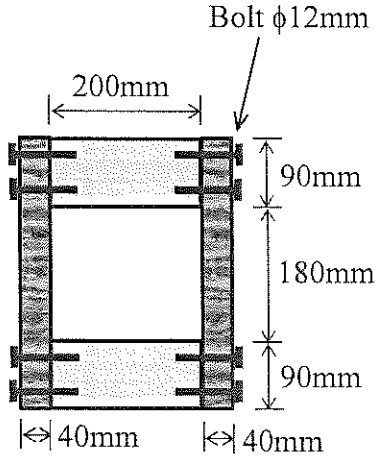


Figure Q3